Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
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ABSTRACT
Large pre-trained language models store substantial amounts of factual knowledge in their model parameters. However, the

1. INTRODUCTION & PROBLEM STATEMENT
Large language models trained purely on text prediction (parametric models) parameterize world knowledge within millions
- Inability to dynamically update knowledge without costly end-to-end retraining.
- Propensity to generate non-factual hallucinations when queried about nuanced factual claims.
- Absence of verifiable provenance: parametric decoders cannot furnish direct source citations or page references for their a

To resolve these deficits, DocMind implements the RAG framework, establishing a hybrid architecture combining parametric

2. SYSTEM ARCHITECTURE & MATHEMATICAL FORMULATION
The RAG pipeline operates via two fundamental components:
- Dense Passage Retriever (DPR): A bi-encoder architecture mapping query x and document chunk z to dense vector embe
  sim(q, d) = <E_q(x), E_d(z)>
  The probability of selecting document chunk z given query x is:
  p(z|x) = exp(sim(q, d)) / sum_j(exp(sim(q, d_j)))

- Parametric Generator: A sequence-to-sequence BART-large model generating target tokens conditioned on the input quer

3. TRAINING, CHUNKING, AND MARGINALIZATION
The document corpus is segmented into non-overlapping or sliding-window overlapping text chunks of 100 to 200 words (ap

Two formulation variants are supported:
- RAG-Sequence: The model selects a single retrieved document chunk to condition the entire generated sequence y:
  p_RAG-Seq(y|x) = sum_{z in Top-k} p(z|x) * prod_{t} p(y_t | x, z, y_{1:t-1})
- RAG-Token: The model marginalizes across different retrieved chunks at each individual token step, allowing multi-docum

4. EXPERIMENTAL DATASETS & BENCHMARKS
We evaluated the framework on standard knowledge-intensive question answering datasets:
- Natural Questions (NQ): Google search queries with human-annotated answers from Wikipedia.
- TriviaQA: 95K question-answer pairs gathered from trivia enthusiasts.
- WebQuestions (WQ): Questions sampled from Google Suggest API.
- CuratedTREC: Information retrieval benchmark for factual question answering.

5. EMPIRICAL RESULTS & METRICS
Evaluation was conducted using Exact Match (EM) accuracy and Token F1-score:
- Natural Questions: RAG achieved 44.5% Exact Match, outperforming closed-book T5-11B (34.5%) by 10.0 points.
- TriviaQA: RAG attained 56.8% Exact Match, surpassing REALM (40.4%) and baseline models.
- WebQuestions: RAG attained 45.2% Exact Match.
Crucially, RAG achieved these results with only 400M parameters, dramatically reducing computational footprint compared

6. CHALLENGES, LIMITATIONS, AND DISCUSSION
Despite strong empirical gains, practical deployment highlights several remaining challenges:
- Retrieval Latency: High latency from vector index lookup and neural encoding during online user queries.
- Chunk Boundary Alignment: Arbitrary chunking can split vital contextual clauses across boundaries.
- Contradictory Evidence: When retrieved passages contain conflicting information, generative models may synthesize blen
- Interpretability and Source Grounding: Verifying that each claim in the generated answer strictly originates from the cited c